When can AI discover adsorption kinetics? A discoverability map

Registered Report, Stage 1: introduction, methods and pilot data

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Abstract

How fast a porous solid takes up a gas or vapour, its uptake rate law, decides how an adsorption separator or a water harvester performs. Equation-discovery methods promise to learn this law, not only its coefficient, from column experiments. The one published test recovered the standard law at a single rate and noise level, with the isotherm supplied exactly. Nobody knows in which regimes such discovery works, or whether it fails where the rate constant stops being identifiable from the data. We propose a pre-registered simulation study on a verified model of a non-isothermal column. It varies how fast uptake is compared with the passage of the adsorption front (the Damköhler number, over four decades), dispersion, noise, error in the supplied isotherm, what is measured (interior probes, or only the outlet), the isotherm's shape, including a stepped, MOF-303-like water isotherm, and the true law. The strongest discovery methods will be compared with the classical identifiability of the rate constant on the same data. Three confirmatory hypotheses rest on interior measurements; for the two that compare boundaries, the pre-declared rules have simulated power of at least 0.95 at 0.4 decade. The outlet-only case is reported descriptively.

1 Introduction

The question

A fixed-bed adsorption experiment reports concentrations and temperatures in time, usually at the column outlet. The quantity an engineer needs from it is the uptake rate law, most often written as the linear driving force (LDF), $\partial q/\partial t = k\,(q^*(c,T) - q)$. AI equation-discovery methods offer to find that law, not only its coefficient. Sparse regression over a library of candidate terms [1, 2], its weak-form [3–5], ensemble [6], Bayesian [7], implicit [8] and exact mixed-integer [9] variants, symbolic regression [10, 11] and neural surrogates embedded in the governing equations [12] all make that offer. This study asks a narrower question, which can be tested:

For which regimes can the adsorption rate law be discovered from data a column experiment can produce, and does discoverability fail at the same boundary as the classical practical identifiability of the rate constant, or earlier?

What is known and what is not new

The physical core of the expected failure is classical and is not claimed here. Near local equilibrium the mass-transfer coefficient is poorly determined by column data. Valocchi derived local-equilibrium validity criteria from temporal moments, in which velocity, dispersion and rate jointly decide whether local equilibrium holds [13]. Hansen and Vesselinov showed that those criteria really test whether mass-transfer-driven dispersion is negligible next to hydrodynamic dispersion [14]. Katsoulas et al. derived asymptotically that, near local equilibrium, finite mass-transfer rates enter only through an apparent dispersion coefficient [15]. In adsorption engineering, Knox et al. found that the LDF coefficient and the axial dispersion can be extracted wrongly from outlet breakthrough, and that interior measurements are needed [16]. Ward and Pini attributed about 70 % of breakthrough output variability to the isotherm parameters [17]. Taylor et al. mapped global parameter sensitivity of sorbing transport across Péclet–Damköhler regimes and found that equilibrium sorption dominates [18]. The counter-evidence is also classical. In the constant-pattern regime, a lumped rate coefficient *is* extracted from frontal breakthrough consistently with shock-layer theory [19]. "k is unidentifiable at high Damköhler number" is therefore not true without conditions. Any boundary must be conditioned on dispersion, isotherm shape, noise and isotherm error, and those are the axes of this design.

The generic failure of sparse regression on such data also has precedents. Single-trajectory identifiability of linear-in-parameter systems, the SINDy model class, depends on the trajectory's geometry; estimation sensitivity is governed by a condition number tied to the data's spatial confinement [20]. For both SINDy and PySR, small eigenvalues of the moment matrix of the candidate functions over the visited states mark combinations the data barely distinguish, and a zero eigenvalue makes distinct equations indistinguishable [21]. SINDy becomes ill-conditioned on slow manifolds, where the recoverable object is the manifold itself [22]. Sparse identification shows noise- and sparsity-induced phase transitions [23]. When trajectories cross time-scale regimes, SINDy fails to find a single global model [24]. Stiffness and timescale separation break rate-law discovery in chemical kinetics [25, 26]. Collinear library functionals make the discovered model non-unique [27]. Three results set general limits. Learning a closed-form model from noisy data has a phase transition: above a noise level, no method can recover the true model [56]. Weak-form SINDy has its own critical noise level [57]. And a system's equations can be uniquely recovered from a trajectory only under strong conditions, which chaos satisfies; non-chaotic dynamics and dynamics with conserved quantities generally do not [66]. Benchmarks on attractor-filling chaotic data, where SINDy's performance showed no significant dependence on scale separation [28], are therefore the easy case. A breakthrough run is the hard one: a transient, non-chaotic single trajectory that settles onto the slow manifold $q \approx q^*(c, T)$, the setting of [20–22]. These results say that discovery must fail

somewhere for such data; none says where, for which methods, or how that compares with what classical identifiability already predicts. The design measures the confinement directly through library conditioning (Methods, §2.7).

Prior equation discovery for adsorption

The closest prior work is Santana et al. [12]. A neural network replaced the solid–fluid mass-transfer term of a fixed-bed model (a universal differential equation, UDE), was trained on outlet breakthrough curves with 5 % Gaussian noise, and was then read out by sparse regression (BIC-selected) and genetic-programming symbolic regression. For the LDF ground truth the recovered form was $p_1 + p_2 q + p_3 q^*$ with $p_2 \approx -0.22$ and $p_3 \approx 0.22$. The isotherm was supplied exactly, and one kinetic rate and one noise level were studied [12]. The same group used a universal ODE on experimental multicomponent fixed-bed data [29] and screened ten hybrid placements of neural networks in chromatography models [30]. Differentiable finite-volume hybrids now learn constitutive laws from transport experiments [31], and nonlocal transport kernels have been learned from sparse breakthrough curves [32]. In surface science, chemical reaction neural networks fitted to transient spectroscopy of CO on Pd(111) produced candidate adsorption mechanisms that fit the same data equally well yet predict distinct coverage dynamics [33]: a learned mechanism, too, can be practically non-identifiable. On the equilibrium side, neural surrogates infer the isotherm from column dynamics [34, 35]. Bootstrap confidence intervals then reveal distinct learned sorption closures with similar predictions, which is practical non-identifiability of a learned closure [36]. Symbolic regression has rediscovered isotherms [37–39, 65]. Kinetic rate-law discovery by symbolic regression with information criteria and design of experiments exists for batch catalysis [40, 41]. Function-level identifiability of UDEs has been separated from parametric identifiability [42], and statistical tests have been used to decide whether isotherm parameters in liquid chromatography must be state-dependent [43]. Three database searches made while compiling our literature review (two OpenAlex queries and one arXiv query), re-run on 12 October 2026 together with seven further queries, found no study that discovers an adsorption rate law from column data other than [12]. This supports, but does not prove, that the regime question is open.

For metal–organic framework (MOF) water sorption the gap is sharper. Water uptake on MOF-303 has been fitted with an LDF model by hand, and the specific diffusion rate falls markedly where the isotherm inflects [44]. Frequency-response measurements explain why: in MOF-303 the water diffusion rate passes through a minimum at the steep step of the isotherm, because the isotherm's slope changes there, not because the molecules move more slowly, and the LDF approximation is least accurate exactly at the step [58, 59]. At high humidity water also condenses into clusters before it diffuses, which a two-concentration model describes better than LDF [60]. For flexible MOFs, the kinetics of adsorption-induced structural transitions were resolved by time-resolved diffraction and a hand-built model, and found to depend strongly on the framework [45]. A loading-dependent $k(q)$ at an isotherm step is therefore real, measured by hand, and an untested target for discovery. A 2026 Perspective on AI for MOF water harvesting does not mention kinetics discovery in its

abstract [46].

What this study adds

Three things are new, if the hypotheses survive. The first is a quantitative discoverability boundary for the strongest available discovery methods, with a mechanism: the realised driving-force fraction against the combined error of the observations and of the isotherm the method is given. The second is a like-for-like comparison of that boundary with profile-likelihood identifiability [47] on the same grid, with the same noise, isotherm error and observation model. The third is a test of whether an errors-in-variables estimator moves the boundary. Isotherm error is an error in a regressor, not in the target, and the study treats it so: the grid will use classical mixed least-squares–total-least-squares fitting [54, 55] inside the exact best-subset search. Errors-in-variables equation-discovery methods [48, 49] will be reported on their own published examples. A negative result on any of these is publishable in the Registered Report format, and the design is built to make it informative. Every null will carry its minimum detectable effect, and every method that cannot be run in its strongest configuration will be reported as such.

Aims and hypotheses

The aim is a map of where the rate law can be discovered, set against where its coefficient can be identified (Fig. 1). Table 1 summarises each hypothesis with its outcome measure, sampling plan, analysis and the interpretation of every outcome. Let δ be the realised driving-force fraction, computed from ground truth as the maximum over the observed (z, t),

$$\delta = \max_{(z,t)\ \text{observed}} |q^*(c, T) - q| / q_{\max}. \tag{1}$$

Let ε_{eff} be the combined relative error of the equilibrium the method is given (isotherm error ε), of the observations (noise σ) and, for O2, of the reconstructed state:

$$\varepsilon_{\text{eff}} = \sqrt{\sigma^2 + \varepsilon^2 + \varepsilon_{\text{rec}}^2}, \tag{2}$$

where $\varepsilon_{\text{rec}} = 0$ for O1. For O2 it is the RMS of $(\hat{q} - q)/q_{\max}$ of the mass-balance-inverted uptake (M1b, Methods) against ground truth. It will be computed once per cell from the noise-free ($\sigma = 0$) O2 observation, so it measures the inversion's structural error; the noise is already counted in σ. Then $R = \delta/\varepsilon_{\text{eff}}$, computed per cell. The rationale is that the rate law's only handle on k is the driving force $q^* - q$. Discovery needs that signal to stand out from the errors that contaminate both its terms: noise on the measured state and error in the isotherm the method is given. R is the simplest dimensionless ratio of the two, a heuristic and not a derived bound. Three of its limits are stated in advance: δ is a maximum over the sampled points, so at high Da it depends on where the samples fall on a steep front; σ is relative to each channel's range while ε is relative to q^*; and the amplification of noise by differentiation is not in R. H2a tests whether R nonetheless collapses the

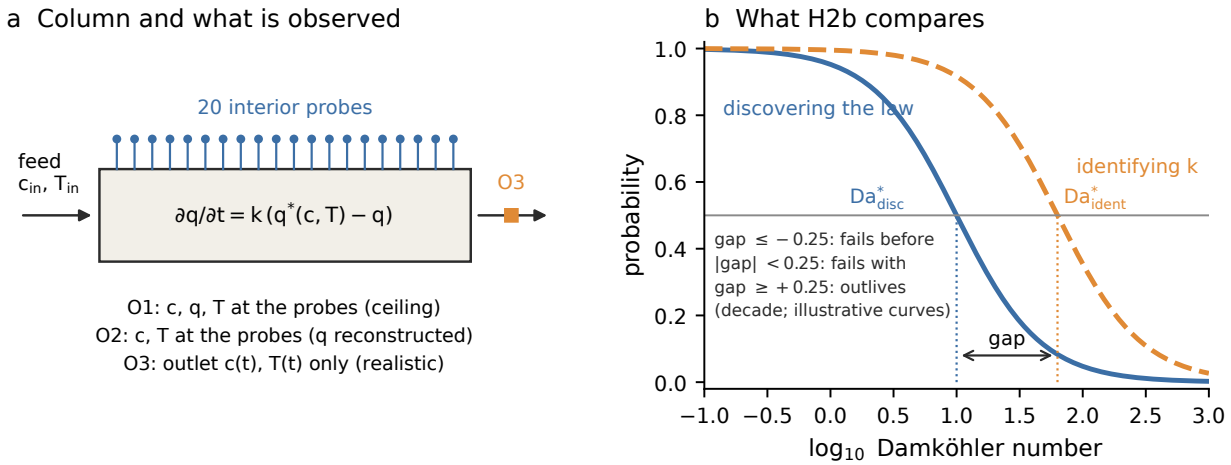


Figure 1: **Study design. a**, The simulated column and the three observation models: O1 and O2 read 20 interior probes (O1 including the solid loading q, O2 without it), O3 only the outlet. **b**, What H2b compares, drawn with illustrative curves (not data): the probability of discovering the rate law and of identifying k given the true law, both against the Damköhler number, their 50 % crossings $\mathrm{Da}^*_{\mathrm{disc}}$ and $\mathrm{Da}^*_{\mathrm{ident}}$, and the pre-declared verdicts set by the gap between them. In this illustration discovery fails before identifiability. `fig_p2_design.py`.

boundary; its falsification would show that it does not.

H2a: the boundary is a driving-force-versus-error boundary. The probability of successful discovery, $P(\text{success})$, collapses onto one curve when plotted against R, across Da, Pe, isotherm shape and observation model (O1 and O2; O3 has no discovery method beyond M1's demonstration). The 50 % crossing lies at R^* with a spread of *less than half a decade* across conditions. *Falsifier:* an R^* spread of one decade or more across conditions. *Pre-declared words:* "*R collapses the boundary*", or "*R does not collapse the boundary; the boundary depends additionally on {named axis}*". A spread of at least half a decade but less than one decade neither confirms nor falsifies H2a and will be reported in the pre-declared words "*inconclusive: R narrows the boundary to a spread of {x} decades*".

H2b: discoverability versus identifiability. For each slice (σ, ε, observation model, isotherm, Pe), we will compare $\mathrm{Da}^*_{\mathrm{disc}}$, the 50 % crossing of $P(\text{success})$ for the best method, with $\mathrm{Da}^*_{\mathrm{ident}}$, the boundary beyond which k is no longer practically identifiable given the *true* law. Both are 50 % crossings from the same logistic estimator (Methods, §2.6). **H2b is confirmatory on O1 only.** There both boundaries rest on 45 estimation replicates per level, and the 0.25-decade decision threshold applies. On O2 and O3, identifiability has 3 replicates on a subgrid, so those comparisons will be reported descriptively, each with its own bootstrap interval, and carry no verdict word. On O1 there are three pre-declared outcomes, reported in these words:

- "*discovery fails before identifiability*": $\mathrm{Da}^*_{\mathrm{disc}} < \mathrm{Da}^*_{\mathrm{ident}}$ by at least the MDE;

- "*discovery fails with identifiability*": the absolute difference is less than the MDE;
- "*discovery outlives identifiability*": $\mathrm{Da}^*_{\mathrm{disc}} > \mathrm{Da}^*_{\mathrm{ident}}$ by at least the MDE. The method then finds the law's form where its coefficient is undetermined.

When either boundary has no crossing inside the tested range, no verdict word will be issued and the range tested will be reported instead (Methods, §2.8).

H2c: errors-in-variables methods move the boundary. The errors-in-variables (EIV) method on the grid (M5, Methods) shifts $\mathrm{Da}^*_{\mathrm{disc}}$ upwards relative to the best non-EIV method at $\varepsilon > 0$; the predicted size is a factor of 2. H2c holds at an ε where the upward shift is detectable, at least 0.25 decade, in at least two thirds of the (isotherm, Pe) slices; the factor 2 grades the size (amendments A1, A2). The comparison is like for like: both methods in the strong form (M5 is strong-form only), on O1. The best non-EIV method over both forms will be reported alongside, descriptively. *Falsifier:* a shift below the MDE at every $\varepsilon > 0$. *Pre-declared words:* "*EIV moves the boundary by {factor}*", or "*EIV does not move the boundary detectably (MDE {x})*". A shift at or above the MDE but below 0.30 decade (a factor of 2) will be reported as "*EIV moves the boundary by {factor}, detectably but by less than the predicted factor 2*", and a shift down of at least the MDE as "*EIV moves the boundary DOWN by {factor}*", a negative result for H2c.

H2d: the MOF case (exploratory, and labelled so). For a step (Type V) isotherm with a rate that drops at the step, as reported for MOF-303 [44], is the state dependence $k(q)$ discoverable, and in which (σ, observation) cells? No directional prediction is made, and no confirmatory claim rests on H2d. Its success criterion is fixed in Methods (§2.5).

Table 1: **Study design table.** Every entry restates the frozen pre-registration (PREREG_P2, commit d562f19, as amended by A1); the power figures are simulated operating characteristics of the rules (pilot P8, `p2_operating_characteristics.py`). The decision threshold is 0.25 decade; MDE in the verdict words means it. Da^*, 50 % crossing of a logistic fit in $\log_{10}\mathrm{Da}$.

Question	Hypothesis	Outcome measure	Sampling plan (N, power)	Analysis plan	Interpretation given to different outcomes
Is the discovery boundary set by the driving force relative to the combined error?	H2a: P(success) against $R = \delta/\varepsilon_{\mathrm{eff}}$ collapses onto one curve; the spread of R^* across slices is below half a decade.	Success of the best method per replicate (exact support, correct signs, coefficient ratio -1 within 10 %); R^* per slice.	60 replicates per cell (15 select the best method, 45 estimate); slices are (Pe, isotherm, observation model), O1 and O2, pooling every σ, ε, Da; cells with R undefined ($\sigma = \varepsilon = 0$ on O1) excluded. Thresholds on the spread (0.5 and 1 decade); simulated verdict correct with probability ≥ 0.997.	Logistic regression of success on $\log_{10} R$ per slice; R^* its 50 % crossing; spread $=$ max $-$ min of $\log_{10} R^*$ with a bootstrap interval; slices without a crossing listed, never imputed.	Spread < 0.5: "*R collapses the boundary*". Spread ≥ 1: "*R does not collapse the boundary; the boundary depends additionally on {named axis}*". Between: "*inconclusive: R narrows the boundary to a spread of {x} decades*".
Does discovery fail before, with or after the rate constant stops being identifiable?	H2b: compares $\mathrm{Da}^*_{\mathrm{disc}}$ (best method) with $\mathrm{Da}^*_{\mathrm{ident}}$ (profile likelihood, true law known); confirmatory on O1.	Gap $\log_{10}\mathrm{Da}^*_{\mathrm{disc}} - \log_{10}\mathrm{Da}^*_{\mathrm{ident}}$ per O1 slice (σ, ε, isotherm, Pe), 96 slices; the hypothesis-level verdict by the two-thirds rule.	9 Da levels $\times$ 45 estimation replicates for both boundaries (15 more select the method). Smallest effect of interest 0.40 decade. Simulated: wrong verdict 0.001 at no gap; correct verdict 0.965 at 0.40 decade and 0.998 at 0.50.	Same logistic estimator for both boundaries, on the estimation set; percentile bootstrap (2000 resamples) within Da levels. Sensitivity: success also requiring $\hat{k} \in [k/2, 2k]$. O2 and O3 (3 replicates) descriptive.	Gap ≤ -0.25: "*discovery fails before identifiability*". $\lvert\mathrm{gap}\rvert < 0.25$: "*discovery fails with identifiability*". Gap $\geq +0.25$: "*discovery outlives identifiability*". No crossing: no verdict. Overall: the word of $\geq 2/3$ of the slices, else "*mixed*".
Does an errors-in-variables fit move the boundary when the isotherm is wrong?	H2c: M5 (EIV best-subset) shifts $\mathrm{Da}^*_{\mathrm{disc}}$ upwards against the best strong-form non-EIV method at $\varepsilon > 0$, on O1; predicted size a factor 2.	Shift in $\log_{10}\mathrm{Da}^*_{\mathrm{disc}}$, M5 minus the best strong-form non-EIV method, per slice (isotherm, Pe, $\varepsilon > 0$) at $\sigma = 2\,\%$, 18 slices.	45 estimation replicates; for $\varepsilon = 2\,\%$ the 9 levels plus 8 refined levels inside ± 0.5 decade of the slice's best non-EIV boundary. Simulated: no false detection; detection 0.997 (refined) and 0.962 (main grid) at 0.40 decade; 0.855 at the predicted factor 2.	Same estimator and bootstrap, on the estimation set; the non-EIV method chosen on the selection set; shift against the best non-EIV method over both forms reported descriptively.	Holds at an ε where the upward shift is ≥ 0.25 in $\geq 2/3$ of its slices. Graded on the median shift: $\geq \log_{10} 2$ (0.30): "*EIV moves the boundary by {factor}*"; 0.25 to $\log_{10} 2$: "*EIV moves the boundary by {factor}, detectably but by less than the predicted factor 2*". Below 0.25 in every slice at every $\varepsilon > 0$: "*EIV does not move the boundary detectably (MDE {x})*". Down by at least 0.25: "*EIV moves the boundary DOWN by {factor}*".
Can a loading-dependent rate at a MOF-like isotherm step be discovered?	H2d (exploratory): no directional prediction.	Discovery of $k(q)$ per cell: support contains q^*_{meas} and q, Spearman $\rho < -0.8$ across 10 bins, top-bin rate ≤ 0.6 of bottom-bin rate.	Step isotherm, true law L2, 60 replicates per cell. No power analysis (exploratory).	Rate of discovery per (σ, observation) cell.	Reported as rates; no confirmatory claim rests on H2d.

2 Methods

Prospective methods are written in the future tense; what has already been done (the ground-truth solves, the harness and its tests, the pilots) is written in the past tense.

2.1 Ground truth

Ground truth comes from the verified finite-difference reference solver of the companion paper [50]. It is a one-dimensional, axially dispersed, non-isothermal column model, and its implementation is verified against closed-form solutions (the companion paper's L0 checks). The uptake law is one of two:

$$\text{L1:}\quad \frac{\partial q}{\partial t} = k\,(q^*(c,T) - q), \tag{3}$$

$$\text{L2:}\quad \frac{\partial q}{\partial t} = k(q)\,(q^*(c,T) - q), \qquad k(q) = k_0\Big[1 - 0.8\,s\Big(\frac{q/q_{\max} - \theta}{0.05}\Big)\Big], \tag{4}$$

where s is the logistic function and θ is the fractional loading at the isotherm's step midpoint, computed from the isotherm at the feed temperature and recorded per solve. Under L2 the rate falls to 20 % of k_0 across the step. L2 is implemented through the solver's state-dependent-rate hook, whose default path is verified unchanged. Non-isothermal physics as in the companion paper is primary; an isothermal control is secondary. There are two isotherms (Table 2). "Langmuir" is the companion paper's default favourable isotherm for a generic sorbent. "Step" is water on a MOF-303-like framework: a cooperative (Type V) isotherm with the cited step position and capacity, the framework's packing, density and heat of adsorption, and a feed at 30 % relative humidity and 30 °C. Column geometry, particle size and flow are the same in both cases. Ground truth is stored with a manifest that records every parameter, the stoichiometric time t_{stoich}, the horizon, and a per-solve breakthrough check (§3); the manifest is public.

2.2 Design axes and sampling plan

Table 3 lists the axes. The primary Damköhler number is $\text{Da} = k\,t_{\text{stoich}}$, the number of kinetic time constants per front passage. The residence-time form kL/u will be reported alongside but is not the axis. In the Langmuir case the interstitial residence time is 0.4 s, against a stoichiometric time of 7200 s (0.35 s against 31 050 s in the step case), so $kL/u = 0.4\,\text{s} \times k$. That is 4×10^{-5} to 4 over the pilot's $k = 10^{-4}$–$10\,\text{s}^{-1}$, and at most about 0.06 over the grid's rate constants (Table 2), so it cannot separate the regimes this study is about (§3).

What has been generated. Noise and isotherm error are applied after the solve, so ground truth needs only the (Da, Pe, isotherm, law) solves: $9 \times 3 \times 3 = 81$, over three (isotherm, law) pairs (Langmuir–L1, step–L1, step–L2). These 81 solves are done. They took 1–414 s each (median 4 s); the slowest were Langmuir solves at $\text{Da} \geq 316$ with the axial dispersion at ×0.2 or ×1. Every solve reached 95 % breakthrough, some after the horizon was extended (§3). This is ground truth,

Table 2: Column and adsorbent parameters of the two ground-truth cases (from the solver's configuration and the public ground-truth manifest).

	Langmuir (generic sorbent)	Step (water on a MOF-303-like framework)
column length; inner diameter	0.10 m; 0.05 m	same
particle diameter; superficial velocity	2 mm; 0.10 m s^{-1}	same
bed porosity; particle density	0.40; 1200 kg m^{-3}	0.35; 1100 kg m^{-3}
axial dispersion D_L at Pe × 1	2.68×10^{-4} m^2 s^{-1}	3.03×10^{-4} m^2 s^{-1}
saturation capacity; heat of adsorption	20 mol kg^{-1}; −50.0 kJ mol^{-1}	25 mol kg^{-1} (0.45 g g^{-1}); −50.5 kJ mol^{-1}
isotherm shape	Langmuir	cooperative step at $P/P_0 = 0.15$ (25 °C)
feed concentration; feed and wall temperature	1 mol m^{-3}; 298 K	0.504 mol m^{-3} (30 % RH); 303.15 K
wall heat-transfer coefficient	10 W m^{-2} K^{-1}	same
stoichiometric time t_{stoich}	7200 s	31 050 s
rate constant k over the 9 Da levels	1.39×10^{-5}–0.139 s^{-1}	3.22×10^{-6}–0.0322 s^{-1}

not a confirmatory result: no discovery method has seen it. (The pre-registration states 3–7 s per solve; the log shows the range above. The figure is descriptive, no part of the design depends on it, and the erratum is recorded in the project's corrections ledger, entry B79.)

Sample and replicates. The unit of analysis is one replicate: one independent (σ, ε) realisation of the observation of one cell. There will be 60 replicates per cell for M2–M6, M9 and the O1 identifiability profile (M8 is not run on the grid; §2.4). Replicates 0–14 form the *selection set*, used only to choose the best method of a slice; replicates 15–59 form the *estimation set*, on which every boundary that enters a verdict is estimated (§2.5). The frozen design had 20 replicates and selected and estimated on the same ones; amendment A1 made both changes before any grid run (§2.11). M6 will run on a separate isothermal control: 9 solves, the 9 Da levels × Pe × 1 × Langmuir, with the energy balance switched off (T held at the feed) and the same horizon rule. These solves are done; all reached 95 % breakthrough. M7, which passed its inclusion gate, needs a Julia search per replicate and so will run on a subgrid, with the same 60 replicates so that it can take part in the selection (the frozen design gave it 3): all Da × $\sigma \in \{0.5\,\%, 2\,\%\}$ × $\varepsilon \in \{0, 2\,\%\}$ × both isotherms × Pe × 1 × O1, the only observation model M7 can see (Table 4). M1 will run only as the minimal demonstration declared in §2.4, nothing more.

Inclusion and exclusion. No replicate, cell or method output will be excluded. A method that raises an error in a cell will be recorded as a failure with the error message, never dropped (§2.5). Cells leave an analysis only where the design already places them outside it: R is undefined at $\sigma = \varepsilon = 0$ on O1 (H2a); O3 enters no H2a slice, because M1's single replicate yields no R^*; O2 and O3 enter H2b descriptively, without verdict words; and no method is run on an observation model it cannot see (Table 4). A boundary that has no crossing inside the tested range is not imputed (§2.8).

Table 3: Design axes (pre-registration §4.2).

axis	levels
$\mathrm{Da} = k\,t_{\mathrm{stoich}}$	9, log-spaced, 10^{-1}–10^{3} (half-decade spacing)
Pe (via the axial-dispersion multiplier)	3: $\times 0.2$, $\times 1$, $\times 5$
noise σ, i.i.d. Gaussian, relative to channel range	0, 0.5 %, 2 %, 5 %
isotherm error ε, smooth multiplicative perturbation of q^*	0, 0.5 %, 2 %, 5 %
observation model	O1 interior c, q, T; O2 interior c, T (q latent); O3 outlet $c(t), T(t)$ only
isotherm	favourable Langmuir; MOF-303-like step
true law	L1 constant k; L2 $k(q)$ (step isotherm only)

Replicates are never re-drawn: every observation is a deterministic function of (cell, replicate, σ, ε, observation model).

Observation model, exactly (`p2_observe.py`). All methods and the identifiability analysis will read data only through one observation function, so every method sees identical corruption. O1 and O2 observe 20 equally spaced interior probes, and O3 observes the outlet; every model samples 200 times uniform over the solved horizon. Noise is i.i.d. Gaussian with standard deviation σ times the channel's range in that cell's ground truth. The measured isotherm is $q^*_{\mathrm{meas}}(c,T) = q^*(c,T)\,[1 + \varepsilon\, g(c/c_{\mathrm{in}})]$, with g a smooth random function of 4 Fourier modes on $[0,1]$, normalised to unit RMS and drawn once per replicate. Every draw is deterministic per (cell, replicate, σ, ε, observation model). The true q^* is never passed to a method.

Settings needed to reproduce the runs. Each observation's random draws come from a generator seeded by the SHA-256 hash of (cell, replicate, σ, ε, observation model), so every replicate can be regenerated on its own. Time derivatives are Savitzky–Golay first derivatives (window 21 samples, polynomial order 3). Sequentially thresholded least squares keeps a standardised coefficient if it is at least 0.05 of the largest, with a ridge of 10^{-6}. The ensemble (M3) fits 100 models, each on 60 % of the rows drawn with replacement, and keeps terms included in at least 60 % of them. The PySR settings of M7 are those of its inclusion gate (§2.4), seeded by the replicate index. Every setting is in the public code, and the versions of all packages are pinned.

2.3 Candidate libraries

The two libraries were fixed before the freeze to the implemented and known-answer-tested harness.

- **Lib-B (equilibrium-informed, primary):** $\{1,\ c,\ q,\ T,\ c\,q,\ q^2,\ c^2,\ q^*_{\mathrm{meas}}\}$. Including q^*_{meas} is legitimate because it is the *measured* isotherm, carrying error ε. The earlier exploratory

script that motivated this study included the exact generating law, which was a defect. An earlier draft of the design specified degree-3 polynomials, $\exp(-\theta/T)$ factors and q^*q products; the tested harness uses the smaller library above. A smaller library makes discovery easier, so the boundaries to be measured are *optimistic with respect to library size*.

- **Lib-A (agnostic):** Lib-B without q^*_{meas}. Under L1 the true law is not representable in Lib-A, because q^* is not polynomial, so Lib-A cannot succeed by the primary metric. It will be reported as held-out fidelity only. The agnostic route to the law's form is M6 (SINDy-PI) on the isothermal control, which finds the implicit Langmuir–LDF structure without any isotherm.

2.4 Methods compared, each in its strongest configuration

The rule is that each method is run in its strongest configuration. A method that cannot be so run will be reported as such and never replaced silently by a weaker one. Implementation choices were fixed against known-answer systems before any grid cell was touched.

M1, the S1 pipeline reproduced. A UDE with a network in the uptake term, trained through the column PDE and read out by sparse and symbolic regression [12]. Its settings are at least those of [12], and stronger where those are weak. Feasibility was resolved by a pilot (§3): M1 will *not be run at the declared scale*. It will run only as a labelled demonstration on a minimal subgrid: Langmuir, Pe × 1, $\sigma = 2\,\%$, $\varepsilon = 0$, O3, the 9 Da levels, one replicate, L-BFGS, at most 50 steps. The O3 comparison rests on the profile likelihood. This is a compute limit, not a finding about UDEs; on a GPU the same code is expected to be feasible.

M1b, mass-balance inversion (O2). From interior $c(z,t)$ the uptake rate follows from the gas-phase balance,

$$\frac{\partial q}{\partial t} = -\frac{\epsilon}{(1-\epsilon)\rho_p}\Big(\frac{\partial c}{\partial t} + u\frac{\partial c}{\partial z} - D_L\frac{\partial^2 c}{\partial z^2}\Big), \tag{5}$$

with q recovered by time integration from $q(z,0) = 0$. This is how an experimentalist with interior probes obtains uptake. It needs no training, and every O1 method will then run on its output, which makes it the strongest O2 competitor. Its coefficients come from the solver's own definition, so the inversion cannot drift from the forward model. Time derivatives use the same Savitzky–Golay rule as O1. Space derivatives use centred finite differences, because the probe spacing is fixed by the observation model and no smoothing window in z may be tuned to the answer. On a fresh solve that is not a grid cell, the inversion recovered q to within the 3 % of $q_{\max}$ bound recorded in the pre-registration; the test measured a maximum interior error of 2.4 % (§2.9).

M2, weak-form SINDy [3, 4]; M2b, its strong-form counterpart. The weak transform uses compactly supported test functions $\phi(t) = ((t-a)(b-t))^8$ and integration by parts, so the target needs no derivative of noisy data. The test-function support will be swept over $\{31, 61, 121\}$

samples and chosen per cell by the held-out weak-system residual on every fifth probe. The law will then be refitted on all probes. A support is admissible only if its training system has at least 3 equations per library term. On the known-answer test, a long support left an underdetermined weak system that fitted a wrong, dense law with zero residual, which a residual criterion would otherwise prefer. M2b is the derivative-based regression on a Savitzky–Golay $\mathrm{d}q/\mathrm{d}t$. Both forms will run every sparse solver (STLSQ, M3, M4).

M3, ensemble or Bayesian SINDy [6, 7]. The inclusion probability of each term will be reported. The implemented configuration is bagged sequentially thresholded least squares with an inclusion threshold of 0.6.

M4, exact best-subset selection by enumeration. This is the MIOSR objective [9], solved exactly because the library is small and so needing no commercial solver. The search is exhaustive over sparsity ≤ 4, BIC selects, and the fit is *unconstrained*. An earlier draft added sign constraints ($k > 0$, zero rate at $q = q^*$). They were dropped before the freeze because "zero rate at $q = q^*$" imposes the very coefficient ratio -1 that the success metric tests, so M4 would succeed partly by construction. Signs are judged by the metric, as for every method. BIC carries a residual-variance floor at $2 \times 10^{-3} \times \mathrm{std}(\mathrm{d}q/\mathrm{d}t)$, tied to the derivative estimator's accuracy at the grid's sampling (RMS 8×10^{-4} at 200 times). At zero noise, plain BIC kept terms of relative size 10^{-4} by fitting the estimator's own systematic error. The floor was first set at 10^{-3} from 400-sample tests, where M6 then bought a spurious q^2 term at zero noise on 200 samples. 2×10^{-3} is the smallest of $\{1, 2, 5\} \times 10^{-3}$ that passes, calibrated on the known-answer system only. These design facts were forced by known-answer tests and declared before the freeze.

M5, errors-in-variables. *On the grid, M5 is EIV best-subset.* It runs M4's exact enumeration, but fits each support by classical mixed least-squares–total-least-squares [55], with the total-least-squares step computed by singular value decomposition [54]. Exact columns (the constant) are fitted by least squares. Error-carrying columns and the target are fitted by total least squares, after each is scaled by its *declared* error standard deviation. For the observed channels this is σ times the channel range. For q^*_{meas} it is $\varepsilon\, q^*$ plus its c-sensitivity. Products get first-order propagation. The target's standard deviation comes from the Savitzky–Golay filter's exact noise gain. Selection uses M4's floored BIC. M5 will run on the full O1 grid, with 60 replicates, in the strong form. Two known-answer tests check it: total least squares removes the attenuation bias that ordinary least squares suffers, and at zero error EIV best-subset recovers L1 with M4's support. M5 is given the true error levels σ and ε, which a practitioner would have to estimate. H2c therefore measures what errors-in-variables fitting gains when the error variances are known: an upper bound on its benefit in practice. The other methods have no use for the error levels and are not given them. Because M5 knows what the identifiability analysis does not, it takes no part in choosing the best method for H2a and H2b (amendment A2); it is used in H2c only, where its advantage is the question.

ODR-BINDy [48] and WENDy-IRLS [49] will be reported on their own published examples only. Both learn *autonomous* ODE systems, in which every state is denoised and given its own equation. The target here is one rate equation with exogenous c and T inside a PDE, and the dominant error is in a regressor, the measured isotherm. Extending either method to that problem would be new method development. This was decided before the freeze. The ODR-BINDy Python port, on the authors' Lorenz example (500 points, 20 % noise), recovered the exact 7-term support. Its relative coefficient error was 5.0×10^{-3}, against about 2.2×10^{-3} read from the paper's heatmap, and it was about 26 times slower (2385 s). Five noise seeds, run before the freeze, settled whether that was scatter or a real gap: the mean relative coefficient error over the seeds was 0.166 against about 2.2×10^{-3} (a gap of about 74), and the exact support was recovered in 1 of 5 seeds. The port will therefore be reported, in the declared words, as "*port, accuracy below the reference*", with that gap stated; M5 on the grid is unaffected. WENDy will be reported as "*NOT RUN*" if no implementation is verified on its authors' own example.

M6, SINDy-PI [8] for rational laws. M6 will be run with Lib-A and *no* measured isotherm, on the isothermal control only (§2.2), because $\exp(-\Delta H/RT)$ is not polynomial. Success is the implicit Langmuir–LDF structure $\{\mathrm{d}q,\ c\,\mathrm{d}q,\ c,\ q,\ c\,q\}$ with correct signs. Its library is $\{1,\ c,\ q,\ c\,q,\ q^2,\ c^2\}$, which is Lib-A without T, because T is constant on the isothermal control. It will run at every σ and at $\varepsilon = 0$ only, because M6 is given no isotherm and isotherm error does not apply. M6 was verified on the known-answer system, with every coefficient (hence b and k) recovered within 5 %.

M7, constrained symbolic regression with PySR [10]. Julia is installed only in an isolated environment. *Inclusion gate, written before any M7 search:* PySR (binary $+, -, \times$; maximum size 12; 40 iterations; PySR's own "best" model selection) was run on the known-answer LDF system ($k = 0.02$; 0.5 % noise on q *and* on the measured q^*_{meas}) for 3 seeds. M7 runs on its subgrid if and only if success holds in at least 2 of 3 seeds, *and* every search finishes within 30 min, *and* peak memory stays under 3 GB. Otherwise it is reported as NOT RUN, with the pilot's numbers, as "infeasible on this machine" or "failing its known-answer test", in those words. *Outcome:* all three seeds recovered the planted law as $k(q^*_{\mathrm{meas}} - q)$ with k within 0.3 % of its true value, in at most 35 s and under 1 GB, so M7 will be run on its subgrid. A first run of the gate recorded "infeasible on this machine"; that was our own wrapper failing (PySR rejected the library's variable names, then a non-serial deterministic setting), and the decision was withdrawn before the freeze. The gate now refuses to decide when a search raises an error, so a software failure cannot again be read as a resource limit.

M8, KAN symbolic extraction [51]. M8 follows the original KAN procedure. It is not a reimplementation of KANDy [52], whose code was not opened. It is a secondary method. *It failed its known-answer test at zero noise in every configuration tried.* The polynomial library at three sparsity penalties routed the law through q^2. Raw variables on one trajectory used q

Table 4: Method–observation applicability (pre-registration §3).

	O1 (c, q, T interior)	O2 (c, T interior)	O3 (outlet only)
M2, M2b, M3, M4, M6[b], M8[a] (regression on q, $\mathrm{d}q/\mathrm{d}t$)	yes	no	no
M5 EIV best-subset (mixed LS–TLS)	yes	no	no
M7 PySR	yes	no	no
M9 slow manifold	yes	no	no
M1b inversion, then every O1 method	—	yes	no
M1 UDE + sparse/symbolic readout	yes	yes	yes
profile likelihood (law known)	ODE per probe	PDE solve	PDE solve

[a] Applicable, but not run on the grid: it failed its cost gate (§2.4).
[b] On the isothermal control only (§2.2).

alone. Raw variables on three trajectories returned $\{1, q^*, {q^*}^2\}$. This is consistent with the reported fragility of KANs on multiplicative terms [53]. M8's grid run was gated on cost. Its best configuration is raw variables with sparsity penalty 10^{-3}. Its subgrid is all Da $\times$ $\sigma \in \{0.5\,\%, 2\,\%\} \times \varepsilon \in \{0, 2\,\%\}$ $\times$ both isotherms $\times$ Pe $\times$ 1 $\times$ O1, with 3 replicates, which is 216 fits. The rule, written into the cost pilot before it ran, was that M8 runs only if one fit at the grid's size takes at most 300 s, so at most 18 h for the subgrid. *The pilot measured 1144 s, so M8 will not be run on the grid.* Over the subgrid that would have been about 69 h. This is a machine limit, and will be reported in those words. It was measured while the machine was shared with other long jobs, which the Stage-2 report will state. On the same known-answer data the timed fit returned only a constant, consistent with the documented failure. M8 will be reported on its known-answer tests only.

M9, slow-manifold discovery [22]. M9 regresses the *state* q, not its rate, on the q-free features $\{1,\ c,\ T,\ c^2,\ q^*_{\mathrm{meas}}\}$ by exact best-subset selection. It is expected to win at high Da by returning the isotherm (plus apparent dispersion). It will be reported as *what is discoverable there*, not as rate-law success. Its own success criterion is given in §2.5.

Which method can see which observation model. This was decided before the freeze; a method is never run where its input does not exist (Table 4). O1 is the optimistic ceiling. O2 and O3 are where the experimentally realistic answer lies, and O3 will be reported first.

2.5 Success metric and secondary accuracy

Success (primary) means that the selected support equals the true law's support *and* every coefficient has the correct sign. For L1 in Lib-B the support must be exactly $\{q^*_{\mathrm{meas}}, q\}$, with coefficient ratio -1 within 10 %. **Accuracy (secondary)** is $|\hat{k} - k|/k \le 10\,\%$, with $\hat{k}$ the negative of the q coefficient. P(success) will be computed per cell over replicates. $\mathrm{Da}^*_{\mathrm{disc}}$ will be obtained from a logistic fit in $\log_{10}\mathrm{Da}$, with a bootstrap confidence interval over replicates (§2.8). A method

that raises an error in a cell will be recorded as a failure with the error message, never dropped.

Success for the other targets, fixed before the freeze:

- *M9:* support $\{q^*_{\mathrm{meas}}\}$ with coefficient 1 within 5 %.
- *L2 (H2d, exploratory), operationally:* with q^*_{meas} held fixed, the fitted law f's implied rate $\hat{k}(q) = -\partial f/\partial q$ is evaluated at the observed samples and averaged in 10 equal bins of $q/q_{\max}$ over the observed range. The state dependence is *discovered* if and only if three conditions hold: the support contains q^*_{meas} and q; $\hat{k}$ decreases across the bins (Spearman $\rho < -0.8$); and $\hat{k}$ in the top bin is at most 0.6 of $\hat{k}$ in the bottom bin. The true k falls to 0.2 of its starting value. This will be reported as a rate per cell, with no boundary hypothesis.
- *Lib-A fidelity:* the relative RMS error of the predicted $\mathrm{d}q/\mathrm{d}t$ (strong form) on the held-out probes (every fifth), per cell.

The best method and its selection. H2a, H2b and H2c use the strongest method of each slice. It is the method with the largest $\mathrm{Da}^*_{\mathrm{disc}}$ on the *selection* replicates (0–14); if no method has a crossing there, the method with the highest mean success. Its boundary is then estimated on the independent *estimation* replicates (15–59), with a percentile bootstrap over those replicates. Choosing the best of several methods and reporting its boundary from the same replicates inflates the boundary: in simulation, by about 0.07 decade for the best of six equally good methods (pilot P8). That is more than a quarter of the decision threshold, and it pushes towards the verdict "*discovery outlives identifiability*". Separating the two sets removes the bias. The identifiability boundary is estimated on the same estimation replicates, so both sides of H2b rest on the same 45. A method with too few replicates in a slice for both sets takes no part in the selection, and M5, which is given the true error levels, takes no part in the selection for H2a and H2b (amendment A2). Every method's own $\mathrm{Da}^*_{\mathrm{disc}}$ will be reported alongside.

2.6 Identifiability criterion

Identifiability will be assessed by the profile likelihood of k [47] with the true law known, at the same σ, ε and observation model. k is *practically identifiable* if and only if the 95 % profile interval (likelihood-ratio threshold 3.84) lies within $[k/2, 2k]$. In O1 the profile integrates the known-law ODE at each probe, driven by observed c and T (monotone cubic interpolation). In O2 and O3 each profile point is a full column solve, scored against the interior c, T probes (O2) or the outlet $c(t)$, $T(t)$ (O3), on a coarse 25-point k grid, with 3 replicates on the declared subgrid. Both PDE profiles are implemented and pass known-answer tests. the coarse neighbours of its interval. It makes up to 6 passes of 41 points (ODE profile) or 21 points (PDE profile), each pass narrowing to the new neighbours. At low noise the interval is narrower than one coarse step, and a single fixed refinement collapsed onto one point below the truth on the known-answer test. The profile is given the *measured* isotherm, so identifiability faces the same isotherm error as discovery.

$\mathrm{Da}^*_{\mathrm{ident}}$ **will be estimated exactly like** $\mathrm{Da}^*_{\mathrm{disc}}$**.** Per replicate, k is identifiable or not under the $[k/2, 2k]$ criterion. P(identifiable) will be fitted against $\log_{10}\mathrm{Da}$ by the same logistic estimator

(§2.8), and $\mathrm{Da}^*_{\mathrm{ident}}$ is its 50 % crossing. The two boundaries compared in H2b are therefore measured by one estimator.

2.7 Mechanistic predictors

For each cell we will record δ from ground truth, the condition number of the library Gram matrix, and its smallest eigenvalue, following [20, 21]. These test whether library conditioning, set by how closely the data hug $q = q^*$, predicts the boundary. They do not enter any verdict.

2.8 Analysis plan

Boundary estimator. $\mathrm{Da}^*_{\mathrm{disc}}$, and R^* under H2a, is the 50 % crossing of an unpenalised maximum-likelihood logistic regression of the replicate-level success outcomes on $\log_{10}\mathrm{Da}$ (or $\log_{10} R$). The confidence interval is a percentile bootstrap (2000 resamples) that resamples replicates *within* each Da level. The design is fixed, and only the stochastic replicates are resampled. No boundary will be reported, and no verdict word issued, when there is no crossing inside the tested range: all successes, all failures, a fitted slope with the wrong sign, or a crossing off the grid. The tested range will be reported instead. A boundary is never extrapolated off the grid.

Calibration, measured before the freeze. On 100 simulated datasets with the design's grid (9 Da levels, 20 replicates) and a known boundary, the estimator had coverage 0.94 (nominal 0.95), bias +0.012 decade, and a standard deviation of the estimate of 0.089 decade.

Decision threshold, smallest effect of interest and power. The verdicts of H2b and H2c compare a gap or shift between two estimated boundaries with a decision threshold of 0.25 decade. The frozen pre-registration called this threshold the MDE, derived as 2.8×0.089 decade, where 0.089 decade is the standard deviation of *one* boundary estimate at 20 replicates. That derivation is right for one boundary against a fixed value, not for the difference of two estimated boundaries. Amendment A1 therefore keeps 0.25 decade as the threshold, names it so, and states power where it belongs: as the probability that each rule returns the correct verdict. The *smallest effect of interest* is 0.40 decade, a factor of 2.5 in Da. It is just under the grid's half-decade spacing, the smallest difference that would change which tested Da level an experimenter should aim at, and, with 45 estimation replicates, the smallest effect the rules detect with probability of at least 0.95. The journal asks for power of at least 0.95.

Operating characteristics of the decision rules (power). We simulated each rule with the frozen ingredients only (pilot P8, §3): planted logistic boundaries as in the calibration (slope 3 per decade), the 9 Da levels, the frozen estimator and thresholds, and 1000 datasets per condition, at the amended 45 estimation replicates and, for comparison, the frozen 20. The two boundaries that H2b and H2c compare were drawn independently, which is conservative if, as expected, estimates made on the same replicates are positively correlated.

- *H2b.* With no true gap, the rule wrongly returns a directional verdict with probability 0.001 (0.054 at 20 replicates). With a true gap of 0.40 decade it returns the correct verdict with probability 0.965 (0.869 at 20), and with 0.50 decade 0.998.

- *H2c.* With no true shift it never reports a detectable shift, on the refined or the main grid. At 0.40 decade it reports a detectable shift with probability 0.997 on the refined grid ($\varepsilon = 2\,\%$) and 0.962 on the main grid ($\varepsilon = 0.5\,\%$, $5\,\%$). At the predicted factor 2 it does so with probability 0.855 on the refined grid.

- *H2a.* Each slice pools many cells, so its R^* is precise. In a simulation of 12 slices of 144 cells (O2's count), the spread verdict was correct with probability of at least 0.997 at true spreads of 0, 0.5, 1.0 and 1.5 decade; with 135 cells (O1's count, the noise-free cells excluded) it was correct in every dataset at 0, 0.25, 0.40, 0.5 and 1.0 decade. Precision does not limit H2a. The simulation spreads the cells' R values evenly, which the real design does only roughly; that is a limit of this check, not of H2a's rule.

- *Selection.* Choosing the best of six equally good methods and estimating its boundary on the same replicates inflated the boundary by 0.071 decade, and the best of nine (about the size of the real pool) by 0.078; selecting on 15 replicates and estimating on the other 45 left -0.001 and $+0.003$ decade.

- *The smallest effect of interest.* At 0.35 decade H2b's power is 0.858, below 0.95; 0.40 decade is the smallest of the simulated effects at which it reaches 0.95.

These figures depend on how sharp the true boundary is. The standard deviation of one boundary estimate at 45 replicates is 0.063 decade at slope 3, 0.091 at slope 1.5 and 0.044 at slope 6. For a shallow boundary (slope 1.5), H2b's power at 0.40 decade is 0.887 and at 0.50 decade 0.982, with a wrong directional verdict at no gap of 0.048; the verdicts will be read with that in mind.

H2c refinement grid. A factor-2 shift is 0.30 decade, close to the decision threshold. H2c will therefore be tested on a refined Da grid of **8 extra levels, log-uniform within** ± 0.5 **decade of the best strong-form non-EIV method's** $\mathrm{Da}^*_{\mathrm{disc}}$ at O1, $\sigma = 2\,\%$, $\varepsilon = 2\,\%$, for each (isotherm, Pe) slice, that method chosen on the selection set and its boundary estimated on the estimation set; no EIV result is read. The 8 levels lie strictly inside ± 0.5 decade, equally spaced in $\log_{10}\mathrm{Da}$ with the two end points excluded. These levels need new ground-truth solves, to be made after the main grid and before any EIV method is run on them. The refinement is part of the design, not a post-hoc rescue.

How each hypothesis will be decided.

H2a. R will be computed per cell. Because σ and ε enter R itself, they are not slicing axes. A slice is one (Pe, isotherm, observation model) combination, pooling every σ, ε and Da in it. P(success) of the best method (§2.5) will be fitted against $\log_{10} R$ by the same logistic

estimator, and R^* is each slice's 50 % crossing. Only O1 and O2 enter: on O3 the sole discovery method is M1's single-replicate demonstration, which yields no R^*. The spread is max − min of $\log_{10} R^*$ over slices, with a bootstrap interval. Slices without a crossing will be listed, never imputed. The verdict words follow the thresholds in the Introduction. When the boundary does not collapse, the axis that separates the R^* values will be named.

H2b. For each O1 slice (σ, ε, isotherm, Pe), 96 in all, the gap is $\log_{10} \mathrm{Da}^*_{\mathrm{disc}} - \log_{10} \mathrm{Da}^*_{\mathrm{ident}}$. Here $\mathrm{Da}^*_{\mathrm{disc}}$ comes from the best method chosen on the selection set and $\mathrm{Da}^*_{\mathrm{ident}}$ from the same logistic estimator, both on the estimation set (§2.5, §2.6). Gap ≤ -0.25 gives "*discovery fails before identifiability*"; gap $\geq +0.25$ gives "*discovery outlives identifiability*"; otherwise "*discovery fails with identifiability*". *The hypothesis-level verdict* is the word held by at least two thirds of the O1 slices that have a verdict; otherwise H2b is reported as "*mixed: the position of discovery relative to identifiability depends on the slice*", with the counts by σ, ε and isotherm. Slices without a crossing are counted and listed, never imputed. O2 and O3 gaps will be reported descriptively, with intervals and no verdict words.

Sensitivity (descriptive). Discovery success requires the law's form and the coefficient ratio, not the value of k, while identifiability concerns k itself. The verdict "*discovery outlives identifiability*" is meant to capture exactly that case: the form is found where the coefficient is undetermined. To show that it is not an artefact of the metric, H2b will also be computed with success additionally requiring $\hat{k} \in [k/2, 2k]$, the identifiability criterion's own tolerance, and reported beside the primary verdict. If the two differ, both are reported and the primary verdict stands.

H2c. H2c will be decided at $\sigma = 2\,\%$ on O1, per slice (isotherm, Pe, $\varepsilon > 0$), 18 in all. In each slice the shift is

$$\log_{10} \mathrm{Da}^*_{\mathrm{disc}}(\mathrm{M5}) - \log_{10} \mathrm{Da}^*_{\mathrm{disc}}(\text{best strong-form non-EIV}),$$

with a bootstrap interval from paired resamples of the two boundaries. It will be estimated on the main grid plus the slice's refined levels for $\varepsilon = 2\,\%$, and on the main grid for $\varepsilon \in \{0.5\,\%, 5\,\%\}$. The shift relative to the best non-EIV method over both forms will be reported alongside, descriptively. Both boundaries are estimated on the estimation set, the non-EIV method chosen on the selection set. The hypothesis *holds at a given* ε if the upward shift is at least the 0.25-decade decision threshold in at least two thirds of that ε's slices with a verdict. The verdict words grade the size of the median shift over those slices, and each slice's own words are reported: at least $\log_{10} 2 = 0.30$ decade gives "*EIV moves the boundary by {factor}*", and between the threshold and 0.30 decade the intermediate words of the Introduction. (The frozen design made the factor 2 itself the criterion; a criterion equal to the predicted effect is met only about half the time when the prediction is exactly right, so amendment A1 moved the criterion to the detection threshold and kept the factor 2 as the graded words.) The hypothesis is falsified if the shift is below the threshold in every slice at every $\varepsilon > 0$.

H2d. Exploratory. We will report, as a rate per cell, in which (σ, observation) cells the loading-dependent rate is discovered under the L2 criterion of §2.5, with no directional prediction and no boundary hypothesis.

Multiple comparisons. The verdicts are threshold rules applied to the point estimates of the boundaries, not significance tests; the bootstrap intervals are reported with every estimate but enter no verdict. The pre-registration specifies no family-wise correction, and none will be applied. Multiplicity is instead controlled by the operating characteristics of the rules: with 45 estimation replicates, a slice with no true gap returns a wrong directional verdict with probability of about 0.001 (pilot P8), so among the 96 O1 slices of H2b about 0.1 spurious verdict is expected, and the hypothesis-level verdict needs two thirds of the slices to agree. Every slice-level verdict will be reported. The best-method selection, the one choice made on the data, is made on replicates that do not enter the estimate (§2.5).

External anchors (exploratory). First, S1 on the map. The Damköhler number of [12] will be recomputed under the definition of §2.2, and also in the residence-time form, and plotted. Our literature review estimated its residence-time value at about 0.9, which is in the kinetically controlled range. If S1 lands in our failure region, the contradiction will be reported and diagnosed, not explained away. Second, real MOF-303 data. The best method per regime will be applied to published MOF-303 water-uptake kinetics [44] and to the digitised breakthrough curve the companion paper already uses. No hypothesis rests on either.

2.9 Outcome-neutral quality checks and positive controls

These checks are orthogonal to the hypotheses: none depends on whether discovery succeeds.

- *Freeze integrity.* The confirmatory grid drivers refuse to run unless the pre-registration is unchanged since the most recent commit titled "FREEZE PREREG_P2". There is no override. A dated amendment is therefore made by committing the amended file under a new such commit; the drivers then check against it, and the repository history keeps every version.

- *Ground-truth validity.* Every solve must reach 95 % breakthrough within its horizon; a solve that does not is extended, never silently truncated. All 81 grid solves and the 9 isothermal-control solves passed (§3). The H2c refinement solves will be held to the same check.

- *Floor and ceiling.* A boundary with all successes, all failures, a slope of the wrong sign or a crossing off the grid yields no verdict word, and the tested range is reported instead (§2.8).

- *Positive control on the grid itself* (amendment A1). In the easiest cells of each isotherm, the lowest Da with no noise and no isotherm error on O1 (all three Pe), the best method other than M5 must recover the law and k must be identifiable in at least 90 % of the estimation replicates. If this fails for an isotherm, no verdict words are issued for that isotherm's slices

(H2a, H2b and its sensitivity analysis, H2c): the failure is reported as the result, and its cause is sought before anything else is interpreted. If one isotherm fails and the other passes, the passing isotherm's verdicts stand, the hypothesis-level verdicts rest on its slices alone, and the withheld isotherm is named.

- *Estimator calibration.* The boundary estimator's coverage (0.94 against a nominal 0.95), bias (+0.012 decade) and standard deviation (0.089 decade) were measured on planted boundaries before the freeze.

- *Information parity.* Every method and the identifiability analysis read data through one observation function; the profile likelihood receives the same measured isotherm as the discovery methods.

- *Positive controls: known-answer tests.* The harness is verified on known-answer systems by a test suite (`tests/test_p2.py`); none of these systems is a grid cell. The components and their checks are:

  - a synthetic LDF series obeying its law to 10^{-3} of the rate scale, and a Savitzky–Golay derivative exact to 10^{-3} on a smooth series;
  - the success metric, which rejects a spurious term of size 10^{-6}, a wrong sign and a ratio error of 50 %, and accepts a 7.5 % ratio error;
  - M4 exact enumeration, STLSQ and the bagged ensemble, each recovering the LDF law at zero noise (M4 with k error below 2 %);
  - the weak form, recovering the law at zero noise and at 2 % noise with k error below 5 %, and beating the strong form at 2 % noise;
  - the O1 runner in both forms with all three sparse solvers;
  - M1b inversion to within 3 % of $q_{\max}$ on a fresh solve, and the O2 pipeline recovering the law when fed the true q. Fed the inverted q at that Da, it does not recover it, a pilot fact declared in §3;
  - the differentiable column used for M1, matching the reference solver to 5×10^{-3} on the exit curve;
  - the O1 profile likelihood (isothermal, and non-isothermal with temperature), the O2 interior-probe profile and the O3 outlet profile, each bracketing k within the pre-registered $[k/2, 2k]$ criterion in the kinetic regime;
  - EIV best-subset: total least squares removing the attenuation bias of ordinary least squares, and recovery of L1 with M4's support at zero error;
  - the L2 success metric and the Lib-A fidelity metric;
  - SINDy-PI recovering the implicit Langmuir–LDF structure with no isotherm supplied, every coefficient within 5 %;

- the boundary estimator's coverage and bias, and the verdict functions for H2a and H2b at the pre-declared thresholds;
- M8's known-answer test, kept as an expected failure that will break the suite if a library change makes it pass.

2.10 Timeline and compute

All computation will run on one CPU workstation without a GPU, one training job at a time, through a job queue with a completion test per job.

- Ground truth: the 81 solves and the 9 isothermal-control solves are complete; the H2c refinement will add 48 solves after the main grid (8 Da levels for each of the 6 (isotherm, Pe) slices).
- O1 and O2 regression methods (M2–M6, M9, and M1b followed by the O1 methods) at 60 replicates over about 2600 cells. These take seconds each. Every method has a grid runner, and every runner refuses to start unless the pre-registration matches its latest freeze commit.
- O1 identifiability: the exact linear-LDF update, about 4.5 s per replicate, so about 65 h at 60 replicates (54 cells $\times$ 16 (σ, ε) $\times$ 60).
- M7 on its O1 subgrid at 60 replicates, and on the refined cells of its Pe $\times$ 1 slices: 5280 searches of at most 35 s each, so at most about 51 h; and the one-replicate M1 demonstration.
- O2 and O3 identifiability: a full column solve per profile point, 3 replicates on the subgrid.

The pre-registration was frozen on 11 October 2026 (commit d562f19), before any discovery method touched a grid cell. The confirmatory grid will start only after in-principle acceptance. We expect to complete it within three months of acceptance and propose to resubmit the Stage-2 manuscript within six months of acceptance. If no journal grants in-principle acceptance within six months of the freeze, the frozen pre-registration provides that the grid runs after a dated amendment saying so, and the study is then reported as a pre-registered study, not as a Registered Report.

2.11 Amendments A1 and A2 to the frozen design

The pre-registration was frozen on 11 October 2026. On 12 October 2026, before any discovery method had run on any grid cell, an internal review written to this journal's Stage-1 criteria found four weaknesses in the frozen analysis plan. The pre-registration's own rule was followed: each fix is a dated, reasoned amendment appended to it (§9 of the pre-registration), committed under a new freeze commit that the grid drivers check. The frozen text is not edited. Amendment A1:

1. raises the replicates from 20 to 60 per cell and separates them into a selection set (0–14) and an estimation set (15–59), because choosing the best method on the replicates that also estimate its boundary biased that boundary upwards (§2.5);

2. declares 0.40 decade as the smallest effect of interest for H2b and H2c and shows by simulation that the rules detect it with probability of at least 0.95 (§2.8);

3. makes H2c hold on a detectable upward shift, keeping the predicted factor 2 as graded words, because a criterion equal to the predicted effect succeeds only half the time when the prediction is exactly right;

4. adds a rule that turns the slice verdicts of H2b into one verdict, an on-grid positive control, and a k-aware sensitivity analysis of H2b; and it runs M7 at the full replicate count so that it can take part in the selection.

A1 also corrects, as errata, statements of fact in the frozen text that no part of the design depends on: the ground-truth solve times, the description of the step case, the bound on kL/u, and the claim that the H2c refinement's end points coincide with main-grid levels (they do only when the boundary falls on a grid level; the 8 levels lie strictly inside ± 0.5 decade either way). No threshold value, axis, method or library changed; A1 adds one verdict word ("*mixed*", for H2b) and changes which outcome counts as H2c holding.

A second review of the amended plan, the same day and still before any grid run, found that the analysis code did not do everything the text said. Amendment A2 (§9 of the pre-registration, under a further freeze commit) therefore:

1. keeps M5 out of the best-method pool of H2a and H2b, because it is given the true error levels that the identifiability analysis is not; M5 serves H2c only;

2. makes Pe a slicing axis of H2b and H2c, as it already was for H2a. The frozen code pooled the three Pe levels into one boundary, a mixture that the power simulation does not describe; per Pe every slice is one boundary over 9 levels, as simulated. It also stops M7, run at $\mathrm{Pe} \times 1$ only, from being compared on different cells;

3. counts a row that raised an error as a failure, as the text always said, where the code had dropped it;

4. adds the outputs the text promised and the code lacked: the bootstrap interval of the H2a spread and of each H2c shift, every method's own boundary, and the H2c shift against the best method over both forms;

5. defines H2c at the hypothesis level over its (isotherm, Pe) slices, and sets the refinement levels per slice;

6. extends the withholding of a failed positive control to the sensitivity analysis, and says what happens when only one isotherm fails: the other isotherm's verdicts stand and the withheld one is named;

7. removes a crash: the best-method choice no longer runs a one-draw bootstrap, which failed when that draw had no crossing.

2.12 Design limitations

- *Synthetic data.* The ground truth is a one-dimensional plug-flow-with-dispersion model. Non-plug flow, which can corrupt the extraction of LDF and dispersion coefficients [16], is outside the model. O3 is the realistic case and will be reported first; O1 is a ceiling.

- *Competitor strength.* The claim is only as strong as the weakest strongest-configuration method. M1 will be run only as a demonstration because of compute. On the grid, M5 is a classical errors-in-variables estimator, not a published EIV discovery method. ODR-BINDy and WENDy are written for autonomous systems and will be reported only on their own examples. M7 passed its inclusion gate. M8 fails its known-answer test and is not run on the grid by its cost gate, measured on a shared machine. Each will be reported in those terms. Lib-B is smaller than first planned, so the boundaries are optimistic with respect to library size. Several methods are not in the competitor set: physics-constrained weak-form symbolic regression [40, 41], partitioning by time-scale regime [24], Bayesian identification by reversible-jump sampling [61], the maximum-likelihood extension of WENDy [67], and parallel symbolic enumeration [62]. "Strongest configuration" therefore means the strongest configuration of each method compared, not the strongest method that exists. Their absence is a limitation, not a finding.

- *The MOF case.* The true law L2 lets the rate fall across the isotherm step and stay low. In MOF-303 the measured rate has a minimum at the step, set by the isotherm's slope, and the LDF form itself is least accurate there [58, 59]. L2 is therefore a simplified, monotone stand-in for that behaviour, and the physically natural target, a rate tied to the isotherm's slope, cannot be expressed in Lib-B. H2d is exploratory for these reasons.

- *Interior measurements.* O1 assumes the solid loading is measured inside the bed. That is demanding but no longer hypothetical: X-ray computed tomography has imaged the adsorbed phase inside a breakthrough column [63]. Designing such experiments for identifiability [64] is outside this study.

- *Axis choice.* Da may not be the right single axis [14, 19]. That is why Pe and isotherm shape are axes and why H2a is stated in terms of R.

- *Sampling.* Stiffness relative to the sampling interval breaks discovery in chemical kinetics [25]. The sampling here is fixed by the observation model, so at the highest Da the kinetic time is short relative to the sampling interval. This will be reported, not varied.

- *Selection of the best method* per slice favours discovery in the sense that the strongest method is used, which is the point. Its statistical cost, a boundary inflated by choosing on the same data, is removed by selecting and estimating on separate replicates (§2.5).

- *Single machine.* The compute limits are those of one CPU. They decide which methods run at scale, not what the methods can do.

2.13 Use of AI tools

The author is the sole author and takes full responsibility for all content. Claude (Anthropic) was used as a tool to write and test code, to help draft and edit the text, and to check the methods and references.

3 Pilot data

Pilot data are excluded from every confirmatory analysis. No discovery method has been run on any grid cell. The pilots fall into three kinds. Generator pilots measured the simulator. Feasibility and verification pilots ran methods only on known-answer systems or on fresh solves that are not grid cells. One pilot (P8) simulated the decision rules alone.

P1, generator pilot. *Measured:* for k on a log grid, the residence-time and run Damköhler numbers, grid convergence ($N_z = 400$ against 800), breakthrough and cost. *Changed:* (i) The primary Damköhler number is $k\,t_{\mathrm{stoich}}$, not kL/u. (ii) At a horizon of $1.5\,t_{\mathrm{stoich}}$ the default column reached only $c/c_{\mathrm{in}} \approx 0.75$ at the outlet. The cause is the thermal wave: the outlet was still about 9 K above feed. At $k = 0.01\,\mathrm{s}^{-1}$ the 95 % crossing was at $2.2\,t_{\mathrm{stoich}}$, and a $4\,t_{\mathrm{stoich}}$ horizon reached $c/c_{\mathrm{in}} = 1.00$. The default horizon is therefore $4\,t_{\mathrm{stoich}}$, subject to a per-solve check that the exit reaches 0.95; a solve that does not is extended, never silently truncated. (iii) The exit curve changed by at most 0.11 % between $N_z = 400$ and 800 at every k tested (10^{-4}–$10\,\mathrm{s}^{-1}$), so $N_z = 400$ is adequate. (iv) The run Damköhler number $k\,t_{\mathrm{final}}$ spans about 1 to 10^5 over that k range. The targeted Da range 10^{-1}–10^3 therefore corresponds to $k \approx 10^{-5}$–$10^{-1}\,\mathrm{s}^{-1}$ at default physics. (v) The residence-time Damköhler number is $0.4\,\mathrm{s} \times k$, from 4×10^{-5} to 4 over the pilot's k range, which is why it is not the axis. The ground-truth solves have since been generated, at 1–414 s each (median 4 s). All 81 reached 95 % breakthrough within their horizon, some after extension to $8\,t_{\mathrm{stoich}}$. Generating ground truth runs no discovery method and was permitted before the freeze.

P2, M1 timing pilot. *Measured* on one cell at Da ≈ 14: the differentiable column matches the reference solver to below 5×10^{-3} on the exit curve, and $N_z = 50$ differs from $N_z = 400$ by 0.55 %. On the study CPU a forward solve takes 22 s, a gradient with respect to k 101 s, and a gradient with respect to a 2×32 network 145 s. That allows about 50 training steps in the 2 h budget, where a UDE needs hundreds to thousands. *Changed:* by the rule declared before the pilot, M1 will not be run at scale (§2.4).

P3, M1b verification. *Measured* on a fresh solve that is not a grid cell: from dense exact c ($N_z = 200$), the inversion recovers q to within 3 % of $q_{\max}$. The median error was 0.013 against a median driving force of 0.18, unchanged at $N_z = 800$, so the error is not numerical diffusion. Fed the true q, the weak-form pipeline recovers the law exactly at Da ≈ 14. Fed the inverted q, it does not; it selects c^2 and T. The inversion error concentrates at the front, where the kinetic signal is. *Changed:*

nothing in the methods or thresholds. The fact is declared because it bears on H2a: for O2, ε_{eff} must include the state-reconstruction error, and δ will be reported against it. The definition of ε_{rec} in the Introduction implements this.

P4, sampling resolution of the O1 profile. *Measured* on a fresh noise-free non-isothermal run: the profile's best k at an interior probe sits $+0.61\,\%$ high with linear interpolation between 200 snapshots, $+0.47\,\%$ with monotone cubic interpolation, and $+0.17\,\%$ at 800 snapshots. *Changed:* monotone cubic interpolation was adopted. The bias is immaterial for the $[k/2, 2k]$ criterion and will be reported with the O1/O2 results. The discovery methods face the same sampling.

P5, known-answer tests that forced design facts. These are M4's BIC residual-variance floor (and its recalibration to 2×10^{-3} at the grid's 200-sample resolution), the adaptive profile-likelihood refinement, M2's admissibility rule of at least 3 equations per library term, the documented M8 failure, and the ODR-BINDy accuracy gap (§2.4). Each was found on a known-answer system and declared before the freeze.

P6, M8 cost gate. *Measured:* one M8 fit at the grid's size took 1144 s, against the 300 s limit written in the pilot script before it ran. The machine was shared with other long jobs at the time. *Changed:* M8 will not be run on the grid (§2.4).

P7, pre-freeze revisions. Drafting this manuscript against the pre-registration and the code exposed places where the two disagreed or where the pre-registration was silent. Each was fixed in the pre-registration, in the open and dated, before any discovery method ran on any grid cell: the text was aligned with the tested harness, quantities that had been named but not defined were defined, and verdict words were added for the gaps between thresholds. The full list is §10 of the pre-registration.

The freeze. The two items that remained have reported: the ODR-BINDy port is reported as "port, accuracy below the reference", and M7 passes its inclusion gate. Both rules were written before the runs that decided them. The pre-registration was frozen by commit d562f19 on 11 October 2026, and amended twice (A1 and A2, 12 October 2026, each under a new freeze commit) before any grid run (§2.11); the grid drivers check the latest freeze commit before they will run, and no discovery method has touched a grid cell. Any later change will be a dated, reasoned amendment, reported as post hoc if made after results exist.

P8, operating characteristics of the decision rules (after the freeze, before any grid run). *Measured* on 12 October 2026, for the journal's power requirement: how often each decision rule returns each verdict when planted boundaries are known, at the frozen 20 replicates and at 40, 45 and 50, and how much choosing the best of several methods on the same replicates inflates its boundary (Methods, power paragraph). It used the frozen estimator and thresholds on simulated

success outcomes only; no ground truth, grid cell or discovery method was involved. *Changed:* it found the frozen design short of the journal's power requirement and the selection biased, and so led to amendment A1: 60 replicates, split 15 for selection and 45 for estimation, which the same simulation shows meets the requirement at the declared smallest effect of interest (§2.11).

Data availability

No confirmatory data exist yet. The ground-truth fields are regenerated deterministically from the stored manifest by `p2_generate.py`, and every corrupted observation is a deterministic function of (cell, replicate, σ, ε, observation model), so every number can be reproduced from the code. The pilot results files (`results/p2_*.json`, `results/odr_bindy_*.json`) and the ground-truth generation log are public in the study repository, https://github.com/abdulhamidbatayhi123/ai-mof-cof-dynamics. At Stage 2 the confirmatory results files, the laboratory log of run dates and the data will be deposited in a public archive with a DOI (Zenodo), time-stamped to show that they were produced after in-principle acceptance.

Code availability

The solver, the harness (`p2/`), the observation model, the grid drivers, the pilots and the tests (`tests/test_p2.py`) are public in the study repository, https://github.com/abdulhamidbatayhi123/ai-mof-cof-dynamics, under the licence given there (MIT). Third-party software used: PySR 2.6.1 [10] (run from an isolated Julia environment) and pykan 0.2.8 [51]; exact versions are pinned in `requirements_p2_venv.txt`. The code state at Stage 2 will be archived on Zenodo with a DOI.

Protocol registration

The pre-registration (`PREREG_P2_discoverability.md`) is public in the study repository and frozen at commit `d562f195ac7e30f0b7abe039946f4f2bd8b3ad6d` (11 October 2026), before any discovery method was run on any grid cell, and amended twice, by A1 and A2 (its §9), under commits whose messages begin "FREEZE PREREG_P2 amendment A1" and "... A2" (12 October 2026, commit 41eac1d for A1), still before any grid run. The file's header still reads "Status: DRAFT, NOT FROZEN", followed by the statement that it is frozen by a commit; it was written before the freeze and left unedited, as A1 records. The freeze is the commit, which the grid drivers verify. After in-principle acceptance, the approved Stage-1 protocol will be registered in a recognised repository (the journal's figshare space, or OSF), and the Stage-2 manuscript will link to it.

Author contributions

A.B. conceived and directed the study and is the sole author. The use of AI tools is described in Methods (§2.13).

Competing interests

The author declares no competing interests.

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